

Flutter

Cross platform for building Mobile Apps

Summer Training • Academic Year 2025 – 2026

Prepared with : Eng.Hadi Heikal



Session 01



01

Setup, Variables & Control Flow

- Introduction to Flutter & Dart
- Tools : VS_Code, Git, GitHub, Figma
- Set up Dart & VS Code
- Data Type & Variable, Naming conventions
- var v.s dynamic
- const v.s final
- User Input
- Conditional Statements
 - (if-else), switch cases, Ternary Operator
- Loops (While Loop, Do While Loop, For Loop)



01

Introduction to Flutter & Dart

What can you do with Programming

- Web Development
- **Mobile App Development**
- Embedded Systems / IOT Systems
- Machine Learning / Deep Learning
- Game Development
- ...

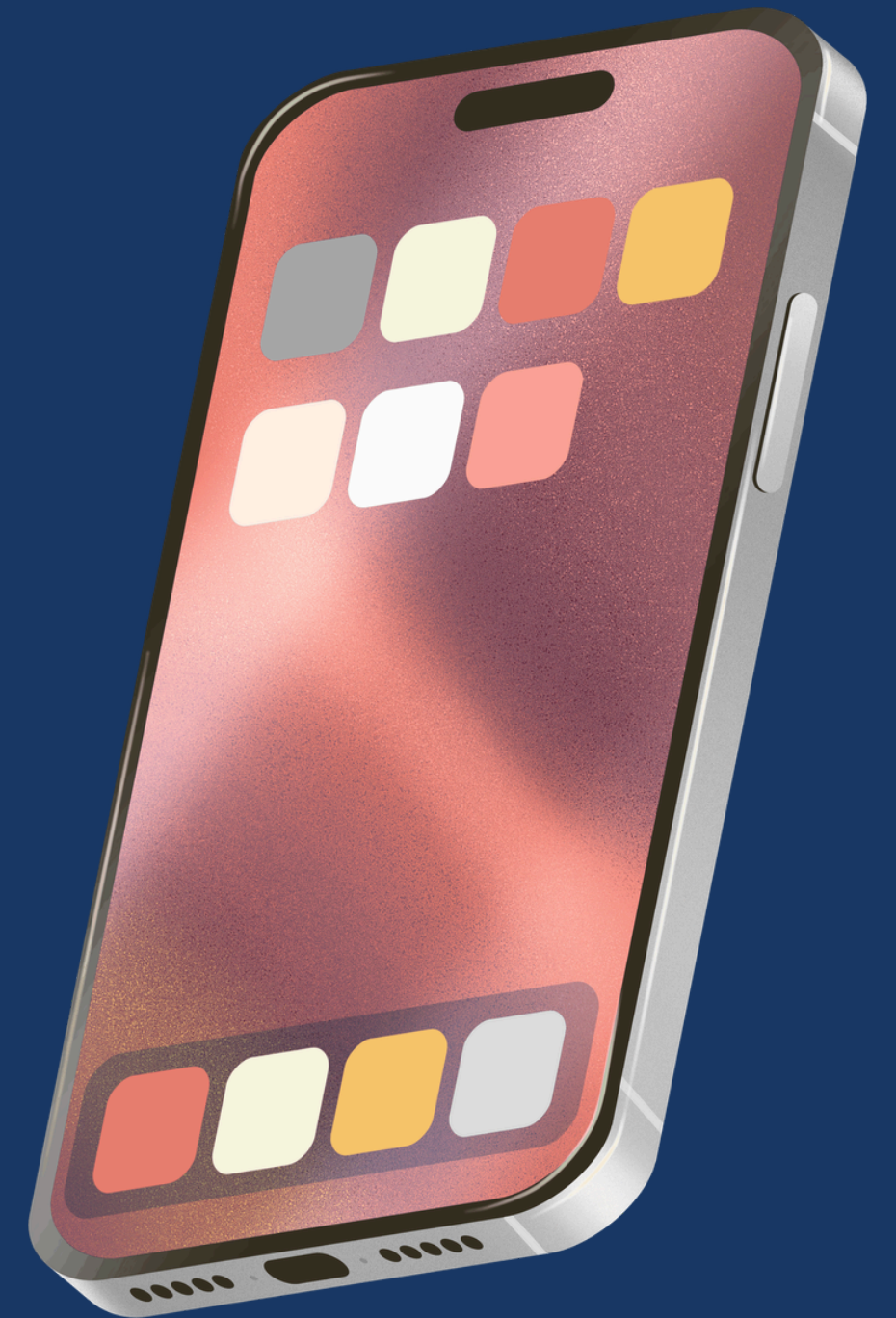


01

Introduction to Flutter & Dart

Mobile development track

Mobile development is the process of making software for smartphones, watches, televisions, and more devices with one of the technologies like Flutter or Native Android/ios.



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Introduction to Flutter & Dart

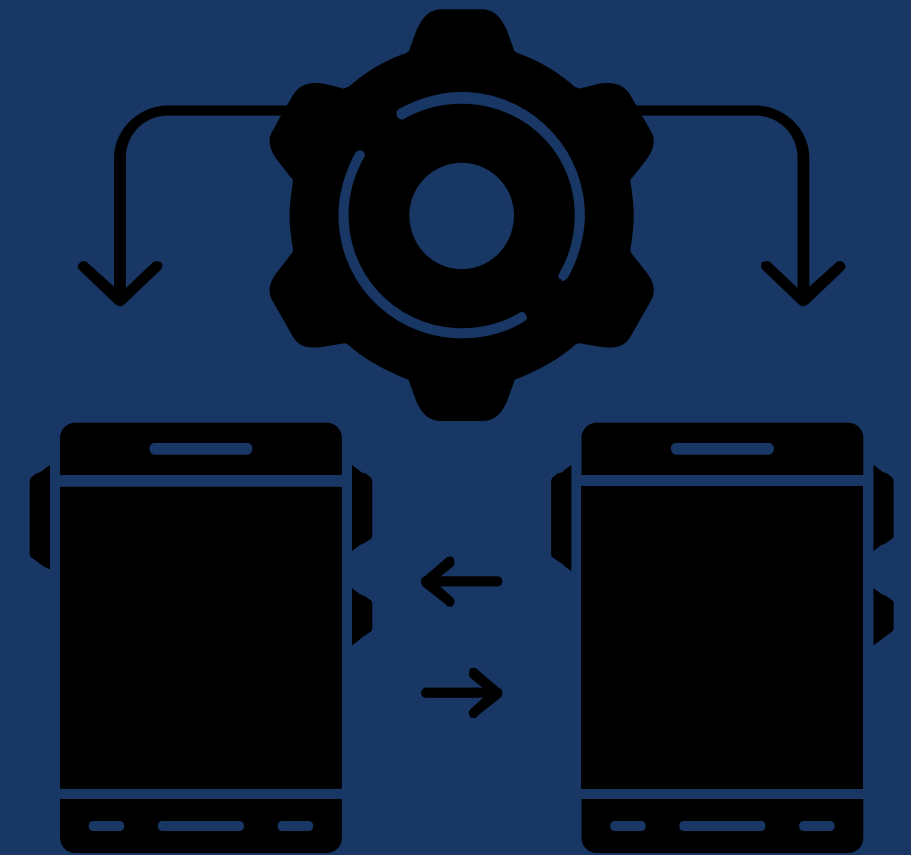
Mobile development approaches

Single Platform

Cross Platform

Native Application

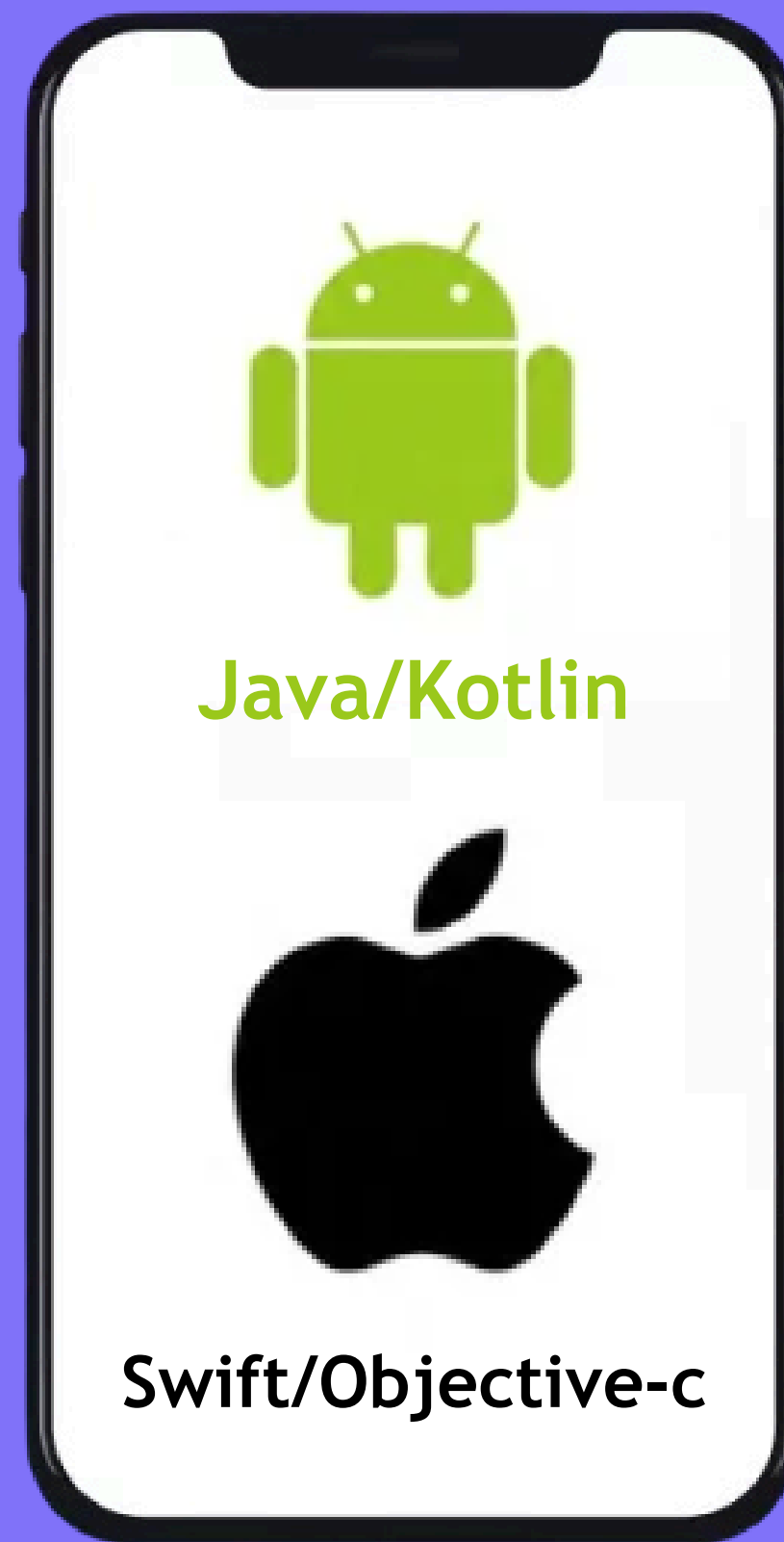
Hybrid Application



Single Platform

VS

Cross Platform



Native



Cross

01

Introduction to Flutter & Dart

What is Flutter?

Flutter is Google's UI toolkit for building beautiful, natively compiled applications for mobile, web, desktop, and embedded devices from a single codebase.

- **Open-Source UI Toolkit:** Built and maintained by Google.
- **Powered by Dart:** Uses the Dart programming language.
- **Cross-Platform (One Codebase):** Builds Android, iOS, Web, & Desktop apps from a single codebase.
- **Established:** First stable release was launched in 2017.



01

Introduction to Flutter & Dart

Flutter advantages

- IOS AND ANDROID APPS, BOTH
- WEB AND DESKTOP TOO
- SINGLE CODE BASE
- COMPILED CODE
- HIGH & FAST PERFORMANCE
- OPEN SOURCE & EASY TO MAINTAIN
- HOT RELOAD AVILABLE
- DEVELOPS NEW FEATURES
- ADVANCED UI
- STRONG COMMUNITY



01

Introduction to Flutter & Dart

Flutter vs React Native vs Native

- FLUTTER: SINGLE CODEBASE, DART, NEAR-NATIVE PERFORMANCE
- REACT NATIVE: JAVASCRIPT, BRIDGE TO NATIVE, HUGE ECOSYSTEM
- NATIVE (SWIFT/KOTLIN): BEST PERFORMANCE, PLATFORM-SPECIFIC, DOUBLE DEV TIME



01

Introduction to Flutter & Dart

Flutter apps in production



Showcase

The world's biggest businesses are building with Flutter. View the showcase and see Flutter apps in production.

Flutterdev



TOYOTA



Google Ads



Alibaba.comTM

Dart language Programming



Dart is a client optimized, garbage collected, OOP language for creating fast apps that run on any platform.

If you are familiar with an object oriented programming language such as Java or C#



Why flutter uses dart?



- Easy to Learn: Dart features a familiar, Object-Oriented syntax that is very similar to Java, C#, and JavaScript, making it easy for beginners and cross-platform developers to pick up quickly.
- Native Performance
- Hot Reload (JIT Compilation): During development,
- Optimized Memory Management





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